



1M1B AI FOR SUSTAINABILITY INTERNSHIP

FoodRescue AI

Smart Surplus Food Redistribution & Hunger Relief System

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IBM SkillsBuild × AICTE × 1M1B

Aligned with UN SDG 2 (Zero Hunger) & SDG 12 (Responsible Consumption)

About the 1M1B Internship

The 1M1B AI for Sustainability Internship, run in collaboration with **IBM SkillsBuild** and **AICTE**, focuses on applying Artificial Intelligence to real-world sustainability challenges.

It builds practical knowledge across AI, sustainability, prompt engineering, IBM tools, and project development — all mapped to the United Nations Sustainable Development Goals (SDGs).



Artificial Intelligence



Sustainability



Prompt Engineering



IBM Tools & Project Dev

Introduction to the Project

Food waste is a major global issue. Restaurants, hotels, hostels, wedding halls, and households often throw away surplus edible food, while many people suffer from hunger.

FoodRescue AI is an AI-powered platform that identifies surplus food, predicts food availability, and connects donors with nearby NGOs and food distribution centers.



Surplus Food



AI Matching



NGOs & Beneficiaries

Problem Statement



Large quantities of edible food are wasted every day.



Many underprivileged people struggle to access nutritious meals.



Lack of coordination between food donors and NGOs.



Food often spoils before reaching people in need.

SDG Alignment

Primarily aligned with SDG 12, while supporting SDG 2 and SDG 11

12 Responsible Consumption & Production

PRIMARY SDG

SDG 12 – Responsible Consumption and Production

- Reduces edible food wastage
- Promotes responsible use of food resources
- Supports sustainable food redistribution

2 Zero Hunger

SECONDARY SDG

SDG 2 – Zero Hunger

- Helps provide food to needy communities
- Connects donors with NGOs for hunger relief
- Improves access to surplus edible food

11 Sustainable Cities & Communities

SUPPORTING SDG

SDG 11 – Sustainable Cities and Communities

- Encourages community-level sustainability
- Strengthens local food support networks

Target Users

FoodRescue AI connects two sides of the same problem — surplus on one end, need on the other.



Restaurants

Daily surplus from kitchens & buffets



Hotels

Banquet & event food left unused



Wedding Halls

Large-scale leftover catering



Households

Everyday excess home-cooked food



NGOs

Last-mile distribution to beneficiaries

GOALS

Project Objective



Reduce food wastage



Support hunger relief efforts



Connect food donors with NGOs efficiently



Use AI to predict surplus food availability



Ensure timely collection and distribution of food

5

Core Objectives Driving the
Platform

Literature Review & Background Study

Studies show that a significant amount of food is wasted globally each year.

Existing food donation systems mainly depend on manual communication, which can cause delays.

AI can improve food redistribution by predicting surplus quantities, matching donors with NGOs, and optimizing collection routes.



Global Scale

Massive yearly food waste despite widespread hunger



Manual Delays

Phone/manual coordination slows donor-NGO matching



AI Opportunity

Prediction, matching & route optimization close the gap

Tools & Technologies Used

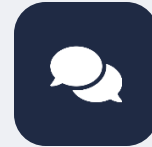
AI TOOLS



IBM Granite AI



IBM Watsonx AI



ChatGPT

TECHNOLOGIES



Python



Machine Learning



Google Forms (Surveys)



Canva (Design)



Google Maps API



End-to-End AI Pipeline

From data collection to AI prediction to last-mile delivery — every layer uses purpose-built tools.

HOW IT WORKS

Step-by-Step Working

1

Food donor registers on the platform

2

Donor uploads details of surplus food

3

AI predicts food quantity and urgency

4

System identifies nearby NGOs

5

Best NGO is matched automatically

6

Collection schedule is generated

7

Food is picked up and distributed

8

Impact data is recorded

System Workflow



A continuous loop: every completed delivery feeds impact data back into the AI model for smarter future matching.

Survey Results & Data Analysis

Survey conducted among students and the public

Q1

Do you think food waste is a serious problem?

Yes  90%

No  10%

90%

said YES

Q2

Would you donate excess food through an app?

Yes  85%

No  15%

85%

said YES

Q3

Do you know any food donation platform?

Yes  30%

No  70%

30%

said YES

Analysis: Most respondents recognize food waste as a major issue and are willing to donate surplus food if a convenient platform is available.

Implementation



Designed the FoodRescue AI concept



Identified target users



Created survey and collected responses



Developed AI-based workflow



Proposed donor–NGO matching system



Designed solution architecture

Impact Created



Social Impact

- Helps reduce hunger
- Supports vulnerable communities



Environmental Impact

- Reduces food waste
- Lowers greenhouse gas emissions



Economic Impact

- Better utilization of food resources
- Reduced disposal costs

Challenges Faced & Solutions

CHALLENGE



Identifying food donors



Quick food collection



NGO coordination



Food spoilage risk



SOLUTION



Awareness campaigns



AI scheduling system



Automated matching



Priority-based collection

Future Scope



Mobile application
development



Real-time tracking
system



Integration with
restaurants &
supermarkets



AI-based demand
forecasting



Expansion to multiple
cities and states

AI Architecture & Feature Mapping

Food Provider ›

Voice / Text Input ›

IBM Granite AI ›

Surplus Prediction ›

RAG: Local Guidelines ›

NGO Matching ›

Route Optimization ›

Impact Dashboard

Feature	AI Tool / Technology
Surplus food prediction	IBM Granite Models, Scikit-learn (Python ML)
AI chatbot for donation queries	IBM Granite, IBM watsonx Assistant
Voice in regional languages	IBM Speech Services, Google Speech APIs
NGO matching & recommendation	Recommendation AI algorithms
Food image recognition (optional)	IBM Granite Vision, Google Vision AI / Teachable Machine
Local donation guidelines	RAG (Retrieval-Augmented Generation) + IBM Granite
Educational tips generation	IBM Granite LLM
Community dashboard analytics	Power BI, Tableau, or IBM Cognos Analytics
Route & pickup optimization	AI optimization algorithms + Maps integration

Responsible AI Considerations



Fair NGO Matching

No bias toward specific NGOs — allocation is based on need, distance & urgency, not favoritism.



Data Privacy

Donor and beneficiary details are never shared beyond what's needed to complete a handoff.



Transparency

Every AI match and prediction can be explained — no black-box decisions.



Human Oversight

AI recommends; NGOs and coordinators retain final say on every match.



Secure Handling

Information is stored and transferred with safeguards against misuse or leaks.

File Edit Selection View Go Run ...

.bob

EXPLORER

.BOB

.vscode

launch.json

ibm project

pycache

food_rescue_ai.cpyt...

example_usage.py

food_rescue_ai.py

QUICK_START.md

README.md

SYSTEM_FLOW.md

settings

custom_modes.yaml

mcp_settings.json

OUTLINE

TIMELINE

food_rescue_ai.py X

food_rescue_ai.cpython-313.pyc

launch.json

README.md

example_usage.py

ibm project > food_rescue_ai.py

```
1 """
2 Food Rescue AI System
3 Takes donor information and suggests nearest NGO with pickup priority
4 """
5
6 import math
7 from datetime import datetime
8 from typing import List, Dict, Tuple
9
10 class NGO:
11     """Represents an NGO with location and capacity"""
12     def __init__(self, name: str, latitude: float, longitude: float,
13                 capacity_kg: float, current_load_kg: float = 0):
14         self.name = name
15         self.latitude = latitude
16         self.longitude = longitude
17         self.capacity_kg = capacity_kg
18         self.current_load_kg = current_load_kg
19
20     def available_capacity(self) -> float:
21         """Returns available capacity in kg"""
22         return self.capacity_kg - self.current_load_kg
23
24     def capacity_percentage(self) -> float:
25         """Returns current capacity usage as percentage"""
26         return (self.current_load_kg / self.capacity_kg) * 100 if self.capacity_kg > 0 else 0
27
28
29 class FoodRescueAI:
30     """Main AI system for food rescue operations"""
31
32     def __init__(self):
```



Hi, I'm Bob

Ask me questions or let me code for you.

[Documentation](#)

Users should independently verify accuracy of AI-generated content.

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What would you like me to help with? context, / for commands, hold shift to drag in

Code

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1

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OUTLINE

TIMELINE

ibm project > README.md > * # Food Rescue AI System

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Food Rescue AI System

A Python-based AI system that helps connect food donors with nearby NGOs for efficient food rescue operations.

Features

- **Smart NGO Matching**: Suggests the nearest NGO based on distance and capacity

- **Priority Scoring**: Calculates pickup priority based on:

- Distance from donor to NGO

- Food quantity vs NGO capacity

- Current NGO load

- **Urgency Classification**: Automatically determines pickup urgency (HIGH/MEDIUM/LOW)

- **Alternative Suggestions**: Provides backup NGO options

How It Works

1. **Input**: Takes donor name, food quantity (kg), and location (latitude/longitude)

2. **Processing**:

- Calculates distance to all NGOs using Haversine formula

- Evaluates each NGO based on capacity and current load

- Computes priority scores (lower = better match)

3. **Output**: Recommends best NGO with pickup priority and alternatives

Usage

Running the Program

```bash

python food\_rescue\_ai.py

```

food_rescue_ai.python-313.pyc

launch.json

README.md

example_usage.py

QUICK_START.md

ibm project > README.md > * # Food Rescue AI System

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OUTLINE

TIMELINE

python-313.pyc

launch.json

README.md

example_usage.py

QUICK_START.md

SYSTEM_FLOW.md

ibm project > SYSTEM_FLOW.md > # Food Rescue AI - System Flow

1 # Food Rescue AI - System Flow

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3 ## Architecture Overview

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7 FOOD RESCUE AI SYSTEM

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11 DONOR

12 INPUT

13

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15 • Donor Name

16 • Food Quantity (kg)

17 • Location (Lat/Long)

18

19 ▼

20

21 FOOD RESCUE AI ENGINE

22

23

24 Step 1: Calculate Distances

25 • Use Haversine formula

26 • Compute distance to each NGO

27

28

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31 Step 2: Evaluate NGO Capacity

32 • Check available capacity

Documentation

Users should independently verify accuracy of AI-generated content.

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What would you like to do with? context, / for commands, hold shift to drag in

Code



FOOD RESCUE AI – EXAMPLE USAGE DEMONSTRATIONS

=====

EXAMPLE 1: Restaurant Donation

=====

Donor: Raj's Restaurant
Food Quantity: 75 kg
Location: (28.6139, 77.209)

RECOMMENDATION:

Recommended NGO: Hope Foundation
Distance: 0.0 km
Pickup Priority: MEDIUM
Priority Score: 13.0

Press Enter to continue to next example...

```
*****  
FOOD RESCUE AI – EXAMPLE USAGE DEMONSTRATIONS  
*****
```

```
=====
```

EXAMPLE 1: Restaurant Donation

```
=====
```

Donor: Raj's Restaurant
Food Quantity: 75 kg
Location: (28.6139, 77.209)

```
-----
```

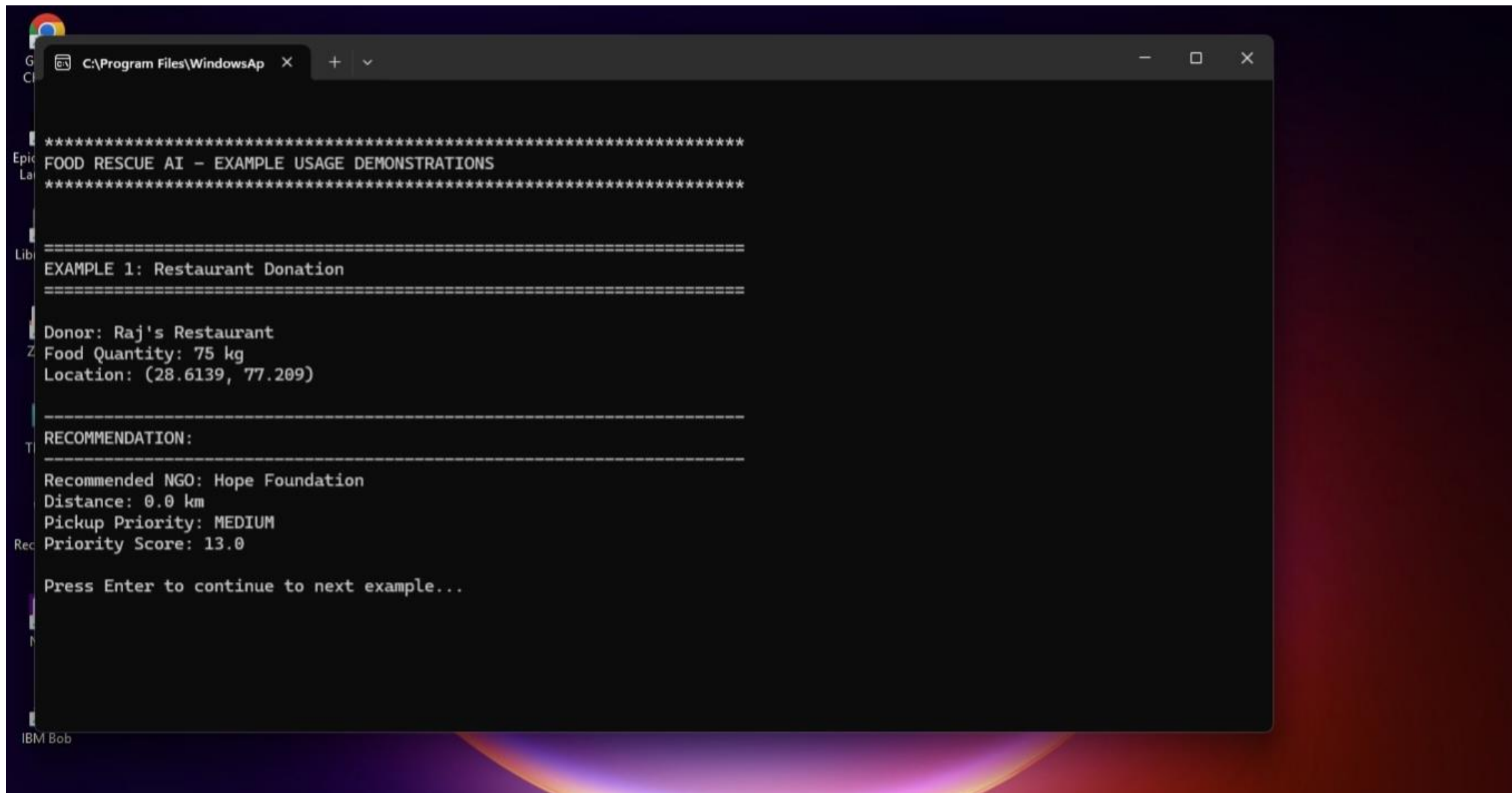
RECOMMENDATION:

```
-----
```

Recommended NGO: Hope Foundation
Distance: 0.0 km
Pickup Priority: MEDIUM
Priority Score: 13.0

Press Enter to continue to next example...


```
C:\Program Files\WindowsAp  X + v - □ ×  
=====  
FOOD RESCUE AI SYSTEM  
=====  
Enter Donor Name: |
```



WRAPPING UP

Conclusion

FoodRescue AI is an innovative AI-powered solution that addresses food waste and hunger simultaneously. By connecting food donors with NGOs and using AI for smart matching and scheduling, surplus food can reach people in need efficiently.

SDG 2

Zero Hunger

SDG 12

Responsible Consumption
& Production



Thank You

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